

Project Design Phase-I

Solution Architecture Date	8 march 2026
Team ID	
Project Name	Project – Quotes Recommendation Chatbot
Maximum Marks	4 Marks

Solution Architecture

Solution architecture is a structured process that bridges the gap between business problems and technology solutions. Its goals are to:

- Identify the best technical solution to address user needs for personalized quotes.
- Describe the structure, characteristics, and behaviour of the chatbot system to stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

System Overview

The **Quotes Recommendation Chatbot** is designed to:

- Understand user queries using **Rasa NLU**.
- Detect user intent (e.g., motivation, love, humor, success).
- Provide personalized quotes that emotionally resonate with the user.
- Support multiple categories: *Inspiration, Motivation, Success, Love, Humor*.
- Enhance user engagement by asking follow-up questions to measure satisfaction.
- Integrate with a **web-based frontend** via REST API for accessibility.

Solution Components

- **Frontend (Web Interface):**
 - HTML/CSS/JS-based interface for user interaction.
- **Backend (Rasa NLU):**
 - Intent recognition and entity extraction.
 - Quote recommendation logic.
 - Emotional state analysis.

- **Database:**
 - Stores categorized quotes.
 - Tracks user preferences and history.
- **APIs:**
 - REST API for communication between frontend and backend.

Data Flow

1. User enters a query (e.g., "I need motivation").
2. Rasa NLU processes input → identifies intent.
3. Backend fetches relevant quotes from database.
4. Chatbot delivers personalized quote to user.
5. Follow-up interaction ensures engagement and satisfaction.